Korea Technology and Information Promotion Agency for SMEs

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Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an Integrated Manufacturing Process Combining Large-Area Aluminum Patterning with AI Monitoring and a High-Corrosion-Resistance Surface Treatment Process**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Machinery and materials		*Information and Communication / Artificial Intelligence*	
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	*Cutting/processing machinery*	*Intelligent Manufacturing Systems*	*AI-based Process Control & Optimization*
Concept of Proposed Technology	**Core Concept** The proposed technology aims to develop an integrated manufacturing process that combines large-area aluminum patterning enhanced by an AI-based sound monitoring system with a high-corrosion-resistance surface treatment process based on Low-Voltage Plasma Electrolytic Oxidation (LV PEO). This integrated approach addresses the growing need for premium home appliance panels requiring sharp, high-end surface textures and machining depth deviations below 50 μm on large aluminum sheets (3×700×2,000 mm). By replacing operator-dependent judgments with quantified AI diagnostics and coupling the patterned surface with an advanced protective LV PEO coating, the integrated manufacturing process delivers both superior functional durability and refined CMF (Color,			

	Material, Finish) quality. **Key Functions and Potential Applications** The Sound Sensor AI Monitoring System analyzes acoustic signatures generated during high-speed machining (50,000 RPM, 6,000 mm/min feed rate) to precisely detect tool wear, vibration anomalies, and cutting-stage transitions. This ensures predictable machining quality and reduces reliance on manual experience. The Robot-Integrated Chip Removal System automatically responds to real-time machining signals, removing accumulated chips immediately to prevent surface defects during long-duration machining of large panels. These intelligent machining functions are combined with LV PEO surface treatment, which provides exceptional corrosion, scratch, and stain resistance while creating a distinct micro-surface tactile texture. The resulting IMP can be applied to premium home appliance panels (refrigerators, washers, air conditioners, styler-type products) and extends naturally into laptops, mobile devices, automotive interior trim, and high-end architectural or interior metal components.
Rationale for Joint R&D Project	**Critical Technical Barriers** Developing this integrated manufacturing process requires overcoming several technical barriers. Maintaining uniform machining quality across large aluminum surfaces under extreme high-speed conditions remains challenging due to micro-level tool wear, breakage, vibration, and surface defects. Extracting accurate diagnostic information from sound-based signals also demands advanced AI models capable of distinguishing subtle frequency shifts associated with tool degradation or variations in surface roughness. Furthermore, integrating AI-driven machining diagnostics with real-time robotic chip removal introduces additional synchronization and reliability challenges that must be addressed at the system level. In addition, ensuring consistent coating quality using Low-Voltage Plasma Electrolytic Oxidation (LV PEO) across large patterned aluminum surfaces requires precise control of electrochemical parameters to produce uniform micro-textures and high corrosion resistance. Addressing these challenges collectively is essential for realizing a stable, scalable, and production-ready integrated manufacturing process. **Why Global Collaboration?** The successful realization of this technology requires the convergence of advanced manufacturing, AI-driven sensing, high-speed machining, and next-generation surface engineering—an area in which Purdue University holds internationally recognized leadership. Purdue's world-class Physical AI research ecosystem includes major centers and laboratories such as the Institute for Control, Optimization and Networks (ICON), the eXcellence in Manufacturing and Operations (XMO) group, and the Assured Autonomy Innovation Institute (A2I2). These organizations are globally known for cutting-edge research in intelligent machining, acoustic signal interpretation, cyber-physical manufacturing systems, and advanced materials processing. Purdue's expertise spans AI-based tool-condition monitoring, sound-signal fusion for process

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	diagnostics, autonomous manufacturing systems, and electrochemical surface engineering—capabilities that are essential for developing the proposed integrated manufacturing process. Korean SMEs, meanwhile, possess deep experience in aluminum machining for premium consumer products and strong pathways to commercialization. Therefore, global joint R&D is critical, as collaboration with Purdue University provides the advanced physical-AI technologies, modeling capabilities, and manufacturing science leadership necessary to successfully develop and validate this next-generation integrated manufacturing process.
Key Words	1. Patterning 2. High-Speed Machining 3. Sound Sensor 4. Plasma Electrolytic Oxidation (PEO) 5. Vacuum Integrity

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)

R&D Objectives

R&D Objectives

The proposed technology aims to develop an integrated manufacturing process that combines large-area aluminum patterning enhanced by an AI-based sound monitoring system with a high-corrosion-resistance surface treatment process based on LV PEO.

Detailed Objectives

- AI Sound Sensing: Machining Intelligence & Data Assetization
  - Filtering out ambient noise and extracting only the machining sounds by applying Sound Sensor AI Monitoring to large-area aluminium patterning (3×700×2,000mm), using an AI algorithm. This establishes a 24-hour autonomous monitoring system that prevents defects such as vibration, chatter, and chip clogging.
  - Converting the "know-how" of skilled technicians into quantitative data assets by performing real-time diagnostics on tool wear and machining conditions.
  - Robot-Integrated Chip Removal: Establishing automation where the "Machining Complete" signal is transmitted in real-time, triggering immediate robot intervention to remove accumulated chips from the large-scale workpiece.
- Advanced Proprietary LV PEO Process (Low-Voltage Plasma Electrolytic Oxidation)
  - Suppressing the arcing and inducing the growth of a highly uniform ceramic coating by controlling plasma at significantly lower voltages (under 250V) compared to standard PEO.
  - High-Aesthetic CMF Finishing: Achieving a sophisticated Matte or Semi-Gloss texture that mitigates the cold sensation typical of metal. It also delivers uniform color consistency (ΔE<1.5) based on global trends (Black and Dark Gray).
  - Maximizing Coating Properties: Achieving superior surface hardness (HV 300–350) and higher corrosion resistance by decreasing the size of the micro-pores.

R&D Target Specifications

Key Performance Indicators	Target value	Measurement and Verification Methods
Surface color deviation (ΔE)	≤1.5	9 points on the surface of the large-area workpiece are measured using a Spectrophotometer
Surface Hardness (HV)	300~350	Micro-Vickers Hardness tester is used to measure surface hardness
Surface roughness (Ra)	≤4	Surface roughness tester measurements (after machining and surface treatment)
Accuracy of machining and tool	≥95%	Comparison of AI sound monitoring system's predictions to actual surface roughness and wear indicators

condition diagnostics		
Film thickness deviation	≤ ±1μm	SEM cross-section measurement and comparison after data-driven control

R&D Plan

- Research Approach & Strategy
  - Through International Joint R&D support, the Korean SME partner will establish an integrated Large-Area Aluminum Texturing and Surface Treatment Technology, leveraging its machining expertise together with an AI-based Sound Sensor Monitoring System to generate a unified, high-fidelity process data stream.
  - Moving beyond simple trial-and-error experiments, Sound Sensor AI models are to be utilized in quantifying how machining conditions impact surface quality. This will enable the proactive prediction and mitigation of potential defects.
  - Initiated by the specific request of S-Company's Home Appliance Design and Component Technology Development teams, the project will adhere to premium appliance quality standards from the early stages. Mass production implementation follows mutual verification of these performance metrics
- Key Tasks & Milestones

Year	Goals	Major Tasks	Milestone
1st	Core Technology optimization	- Advanced Sound Sensor AI algorithms dedicated to hexagonal edges (120 degrees) - Establishment of basic process conditions for LV PEO	Achievement of cutting depth deviation of less than 50μm
2nd	Platform integration and verification	- Establishment of a cyber-physical model for processing and surface treatment integration - Production of large-area mock-ups for home appliances	Securing 90% first-time pass rate (FPY)
3rd	Mass production securing and verification	- Global Standard quality certification (corrosion resistance and weather resistance testing) - Manufacturing cost innovation model verification and GVC order acquisition activities	Final Yield Rate: 97% or Higher

- Project timetable

Research activity name	1st Year (Q1~Q4)				2nd Year (Q1~Q4)				3rd Year (Q1~Q4)			
Sound Sensor AI algorithm development and simulation												
LV PEO low-voltage stabilization process design												

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Building a data platform for machining and surface treatment integration												
Home appliance prototype production and evaluation												
Global Quality Certification and Procurement Contracts												

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	● Development of Algorithms & Design of AI-based Monitoring ■ Technical Contribution & Research Scope · Development of numerical models that utilize high-frequency vibrations to distribute cutting loads during aluminum machining, in order to maximize surface Integrity. · Establishing a virtual machining environment to predict plastic deformation at 120° obtuse-angle edges and to maintain a machining /cutting depth deviation of 50μm or less. ■ Specific Tasks · Designing data driven protocols that analyze sensor data during machining and transmit optimized parameters to the high-speed machining equipment in real-time. · Conducting precision analysis of surface roughness and microstructural changes immediately after machining using advanced equipment such as SEM (Scanning Electron Microscopy) and AFM (Atomic Force Microscopy) to provide guidelines for optimal machining conditions.
Expected R&R of Korean SME Partner	● Development of Integrated System & Testing ■ Technical Contribution & Development Tasks · Enhancing production efficiency and securing a competitive edge by integrating pre-existing machining techniques with Sound Sensor AI Monitoring and Robot-Integrated Chip Removal. · Implementing LV PEO and Soft-Sparking processes to optimize surface coatings. This ensures high corrosion resistance on precision-machined surfaces while eliminating edge burning (surface defects at the corners). ■ Testing & Validation · Conducting performance validation tests according to the quality standards of global appliance leader S-Company, targeting: corrosion resistance (72 hours of Salt Spray Test), surface hardness (HV350) and color uniformity (ΔE) ■ Collaboration & Resource Allocation · Establishing a collaborative framework to share data generated from actual mass-production samples with research institute to improve system precision. · Deploying PEO facilities, Low-Voltage Pulse Power Supply units, quality measurement instruments, and a team of specialized technical experts.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	**1. Technical Achievements** ● Establishment of an Intelligent Integrated Quality Control System ■ By integrating the Sound Sensor AI Monitoring and VAM physical parameters acquired through international joint R&D, we have established a large-area aluminium patterning texture technology. This creates a seamless manufacturing process linked directly with LV PEO. ■ The Sound AI technology ensures surface integrity throughout the entire manufacturing lifecycle, allowing for the proactive prediction and control of process errors. ● Ultra-Precision Shape Retention & Elimination of Edge Defects ■ By controlling burr height to under 5μm immediately after machining, the process achieves both ultra-precision shape retention—even in complex hexagon patterns—and high-aesthetic exterior quality. ● Securing Global Reliability for Domestic Surface Treatment Technology ■ By securing superior hardness (HV350) and corrosion resistance (72 hours) compared to standard anodizing, we will have realized high-value-added material and component technology. **2. Commercialization Potential** ● Securing a Position in the Global Premium Home Appliance Value Chain ■ Successfully entering the global value chain as a direct supplier for S Electronics' premium lineups (e.g., Bespoke, Signature). ■ Securing a competitive advantage by providing "Super-Gap" CMF (Color, Material, Finish) solutions that meet the rigorous aesthetic and tactile demands of the high-end audio market (e.g., Bang & Olufsen). ● New Market Development via Technology Scalability ■ Leveraging unrivaled ceramic-like textures and uniform "Deep Color" technology to expand into high-tech industries which require extreme durability and reliability, such as mobile devices, aerospace, and medical equipment housings. **3. Industrial and Economic Impact** ● Optimizing Cost Structure through Manufacturing Innovation ■ By merging existing machining expertise with Sound Sensor AI Monitoring System, securing a production yield of 97% or higher is possible. The establishment of IMP enables significant cost optimization. ● Energy Savings and Carbon-Neutral Manufacturing (ESG Management) ■ Operating at low voltage levels (under 250V) compared to conventional high-voltage processes (600–800V) will result in reduction in power consumption by more than 70%. ■ Contributing to national carbon neutrality goals by optimizing energy efficiency and adopting eco-friendly manufacturing processes. ● Securing Global Leadership of Strategic National Technology ■ Through international joint R&D, the Korean SME aims to secure original machining

	algorithms and practical technical partnerships, elevating the technological standard of the global home appliance component sector to a world-leading level.

6. References

(1) Choi, W., Kim, H. Y., Jeon, J. W., Chang, W. S., & Cho, S. H. (2017). Vibration-assisted femtosecond laser drilling with controllable taper angles for AMOLED fine metal mask fabrication. *Materials*, *10*(2), 212. https://doi.org/10.3390/ma10020212

(2) Dehnavi, V., Luan, B. L., Liu, X. Y., Shoesmith, D. W., & Rohani, S. (2015). Correlation between plasma electrolytic oxidation treatment stages and coating microstructure on aluminum under unipolar pulsed DC mode. *Surface and Coatings Technology*, *269*, 91-99. https://doi.org/10.1016/j.surfcoat.2014.11.006

(3) Sim, Y., Kim, E., Lee, H., Sohn, J., & Jun, M. B. G. (2025). Structured sound data augmentation based on spectral analysis for machine tool operational state monitoring. The International Journal of Advanced Manufacturing Technology, 1-15.

(4) Han, C., Kang, Y. S., Sim, Y., Park, H. W., & Jun, M. B. G. (2025). LISTEN: Lightweight Industrial Sound-representable Transformer for Edge Notification. arXiv preprint arXiv:2507.07879.

(5) Danil Yu Pimenov, Maitri Kiran, Navneet Khanna, Giuseppe Pintaude, Marina Cardozo Vasco, Leonardo Rosa Ribeiro da Silva & Khaled Giasin (2023). Review of improvement of machinability and surface integrity in machining on aluminum alloys https://link.springer.com/article/10.1007/s00170-023-12630-4

(6) Duc-Thuan Nguyen, Jong-Myon Kim (2025). Tool wear detection using novel acoustic emission features and a two-stage Mann-Whitney *U* test https://www.sciencedirect.com/science/article/abs/pii/S0003682X25004244

(7) Lei Yang, Guochao, Da Xie, Anand Kumar, Li Sun, Xinhang Shang, and Jun Wu (2025). Acoustic mechanism analysis of milling process through workpiece and spindle vibrations https://journals.sagepub.com/doi/abs/10.1177/10775463251355358

(8) Fengyan Hou, Rukmini Gorthy, Ian Mardon, Da Tang & Chris Goode (2022). Low voltage environmentally friendly plasma electrolytic oxidation process for titanium alloys https://www.researchgate.net/publication/359879982_Low_voltage_environmentally_friendly_plasma_electrolytic_oxidation_process_for_titanium_alloys

(9) Viswanathan S. Saji (2025) Electrolyte and electrolyte-additives for improved plasma electrolytic oxidation on magnesium alloys https://www.sciencedirect.com/science/article/pii/S2666523925000091

(10) Mordor Intelligence: Smart Home Appliances Market Size & Share Analysis - Growth Trends and Forecast (2026 - 2031) *https://www.mordorintelligence.com/industry-reports/smart-home-appliances-market*